

Forecasting Weekly 10-Year Government Bond Yield Changes

Using JPMaQS Macro-Quantamental Indicators

BFI JPMaQS Macro-Quantamental Data Competition | Yun Liu | University of Chicago

1. Economic Rationale and Predictor Selection

The standard decomposition of nominal bond yields into a real rate and an inflation compensation component (the Fisher identity) motivates the three principal prediction channels used in this model: inflation expectations, monetary policy stance, and risk appetite. Weekly yield changes are inherently noisy — they blend permanent macro revisions with transient liquidity and flow effects — but point-in-time macro-quantamental indicators provide the cleanest available signal because they reflect exactly the information set available to market participants at each date, free of look-ahead revision bias.

Inflation and Real Yield Signals

Realized CPI inflation (CPIH_SA_P1M1ML12, CPIC_SA_P1M1ML12) and 1-year-ahead inflation expectations (INFE1Y_JA) are the most direct bond yield drivers: higher realized and expected inflation mechanically raises the compensation investors demand, and signals future central bank tightening. An engineered real-yield spread feature (nominal yield carry minus expected CPI) captures the valuation wedge: compressed real yields historically precede yield normalization. This is consistent with the growing empirical literature linking real rate mean-reversion to subsequent yield rises (e.g., Campbell & Shiller, 1991; Cochrane & Piazzesi, 2005).

Bond Carry and Yield Curve Shape

Bond carry — the return from holding a bond assuming an unchanged yield curve — at 2-year (DU02YCRY_NSA) and 5-year (DU05YCRY_NSA) maturities encodes current yield curve slope and curvature. Positive carry is a documented predictor of near-term bond returns across DM markets (Kojien et al., 2016). An engineered carry-slope feature (5Y carry minus 2Y carry) additionally captures the curvature of expectations about the policy rate path: a flattening carry slope often signals an impending policy pivot, creating asymmetric yield dynamics across the curve.

Risk Appetite and Cross-Asset Signals

Equity excess returns (EQXR_NSA) emerged as the single strongest predictor in feature importance analysis. This reflects the well-documented risk-on/risk-off dynamic: equity rallies reduce safe-haven demand for government bonds, pushing yields higher. The cross-sectional z-score of equity returns (EQXR_NSA_csz) captures relative risk appetite across countries, allowing the model to distinguish country-specific from global equity-bond co-movement.

Credit and Growth Conditions

Credit-to-GDP (PCREDITGDP_SA) proxies for above-potential economic expansion: rapid credit growth signals overheating, raising expected policy rates and term premia. This variable has historically led yield rises by 1–2 quarters in panel studies of DM economies.

A pooled panel of nine developed market countries (USD, EUR, GBP, JPY, AUD, CAD, CHF, SEK, NOK) was used rather than country-specific models. This exploits the common macro-yield transmission mechanism across DM economies while country dummy variables absorb persistent structural differences (e.g., JPY's chronic near-zero-rate regime, CHF's safe-haven premium).

2. Model Architecture and Training Methodology

Ensemble Design

A VotingRegressor ensemble combines three base learners with complementary inductive biases, equal-weighted to minimise variance relative to any single model:

- **Ridge.** Ridge Regression ($\alpha = 194$, selected by 5-fold CV): An L2-penalized linear model that preserves all predictors but shrinks correlated macro indicators toward zero jointly. Its standardized coefficients are directly interpretable as signal strengths and serve as the economic accountability layer of the ensemble.
- **ElasticNet.** ElasticNet (α , l1_ratio searched jointly): Combines L1 sparsity with L2 grouping, implicitly selecting the most informative macro signals while handling the high pairwise correlations among inflation and carry indicators.
- **HistGradBoost.** Histogram Gradient Boosting: Captures non-linear regime interactions — for instance, inflation signals should only predict yield rises when the central bank is not constrained at the zero lower bound. Handles missing values natively, which is essential because many JPMaQS indicators are published monthly or quarterly and contain structural gaps.

Cross-Validation and Temporal Discipline

Walk-forward TimeSeriesSplit with 5 folds and a 4-week gap was applied to the pooled panel (9,486 training rows). The gap prevents leakage from macro releases whose information state spans the fold boundary. Panels were sorted by date before splitting — not by country — to ensure strictly chronological folds that respect the causal structure of macro-financial data. All hyperparameters (Ridge α , ElasticNet α and l1_ratio) were selected via nested CV on the training set only.

Feature Engineering

Three engineered features were added beyond the 30 downloaded JPMaQS indicators:

- `carry_slope` = `DU05YCRY_NSA` – `DU02YCRY_NSA`: captures yield curve carry steepness.
- `ryld_spread` = nominal 10Y yield carry minus expected CPI: real yield valuation signal.
- `EQXR_NSA_csz`: cross-sectional z-score of equity excess returns across the 9 countries.

All features were forward-filled across short gaps (≤ 4 weeks) to handle monthly/quarterly release schedules, then median-imputed within each pipeline to handle any residual missing values robustly. StandardScaler was applied before Ridge and ElasticNet; HistGradBoost is scale-invariant.

3. Results and Key Findings

Cross-Validation Performance

The ensemble achieved a mean CV RMSE of 0.927% per week (± 0.163 across folds), outperforming each constituent: Ridge 0.959%, ElasticNet 0.975%, HistGradBoost 0.937%. Fold-level RMSE ranged from 0.741% to 1.198%, reflecting the varying macro volatility across sub-periods — notably, the highest-error fold coincides with the late-2024 repricing of U.S. rate cut expectations. The ensemble's lower variance across folds (± 0.163 vs. ± 0.187 for Ridge) confirms the diversification benefit of combining linear and non-linear learners.

Out-of-Sample Performance (52 weeks, March 2025 – March 2026)

The ensemble achieved an OOS RMSE of 0.688% per week — materially lower than the CV estimate of 0.927%, indicating that the model generalizes well to the OOS regime rather than overfitting to the training sample. OOS directional accuracy reached 57.9%, compared to 54.5% on the training out-of-fold sample and a 50% naive baseline. A directional accuracy above 55% is practically meaningful: even modest directional edge translates into positive risk-adjusted returns in a long-short duration strategy after transaction costs at weekly rebalancing frequency.

Feature Importance and Economic Interpretation

Ridge coefficient magnitudes on standardized features (Figure 1) rank the predictors in order of economic signal strength. The top five — equity excess return (EQXR_NSA, 0.122), 5Y bond carry (DU05YCRY_NSA, 0.099), 2Y bond carry (DU02YCRY_NSA, 0.070), engineered carry slope (0.070), and cross-sectional equity z-score (EQXR_NSA_csz, 0.062) — all align with the economic channels described in Section 1. The top CPI predictor (CPIH_SA_P1M1ML12, 0.060) and inflation expectations (INFE1Y_JA, 0.046) round out the inflation channel. Country dummies (CHF, EUR) absorb persistent structural differences rather than acting as information-bearing predictors.

Country-Level Accuracy

CHF (57.7%) and JPY (57.5%) achieved the highest directional accuracy. Both are safe-haven currencies whose bond yields exhibit tighter links to global risk appetite — the equity-yield channel dominant in EQXR_NSA — consistent with the feature importance results. USD showed the lowest directional accuracy (52.6%), reflecting the greater idiosyncratic policy noise in U.S. Treasuries during the OOS period of recalibrating Fed rate expectations.

Model Limitations

Out-of-fold residuals show a modest negative mean across all countries (−0.12 to −0.35% per week), indicating the model systematically under-predicts return magnitudes. This is a known consequence of L2 shrinkage: Ridge and ElasticNet compress predictions toward zero, reducing tail accuracy while improving average error. Future work could correct for this bias with a post-hoc scaling factor or by incorporating a heavier-tailed loss function. Additionally, the model uses a static feature set; a regime-switching extension that up-weights recent observations during high-volatility periods could further improve OOS robustness.